

CLOUD COMPUTING CONCEPTS

with Indranil Gupta (Indy)

ORIENTATION TOWARDS
CLOUD COMPUTING
CONCEPTS:

SOME BASIC
COMPUTER SCIENCE
FUNDAMENTALS

WHAT THIS LECTURE IS ABOUT

- Covers basic concepts in Computer Science that will be assumed in the Cloud Computing Concepts (C3) course
- For those of you already familiar, it's a refresher
- Use this as reference if you don't understand (during the course) how the basics are being used

WHAT'S IN THIS LECTURE

- I. Basic datastructures
- II. Processes
- III. Computer Architecture
- IV. $O()$ notation
- V. Basic probability
- VI. Miscellaneous

I. BASIC DATASTRUCTURES: QUEUE

- Queue: First-in First-out datastructure

Remove from head $\leftarrow$

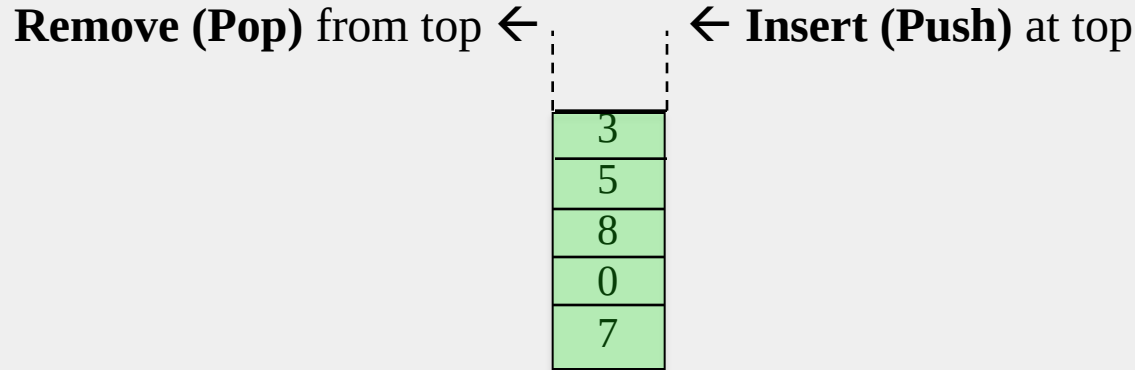
3	5	8	0	7
---	---	---	---	---

 $\leftarrow$ **Insert** at tail

- Next item dequeued (removed) is 3.
 - Then 5
 - Then 8
 - And so on

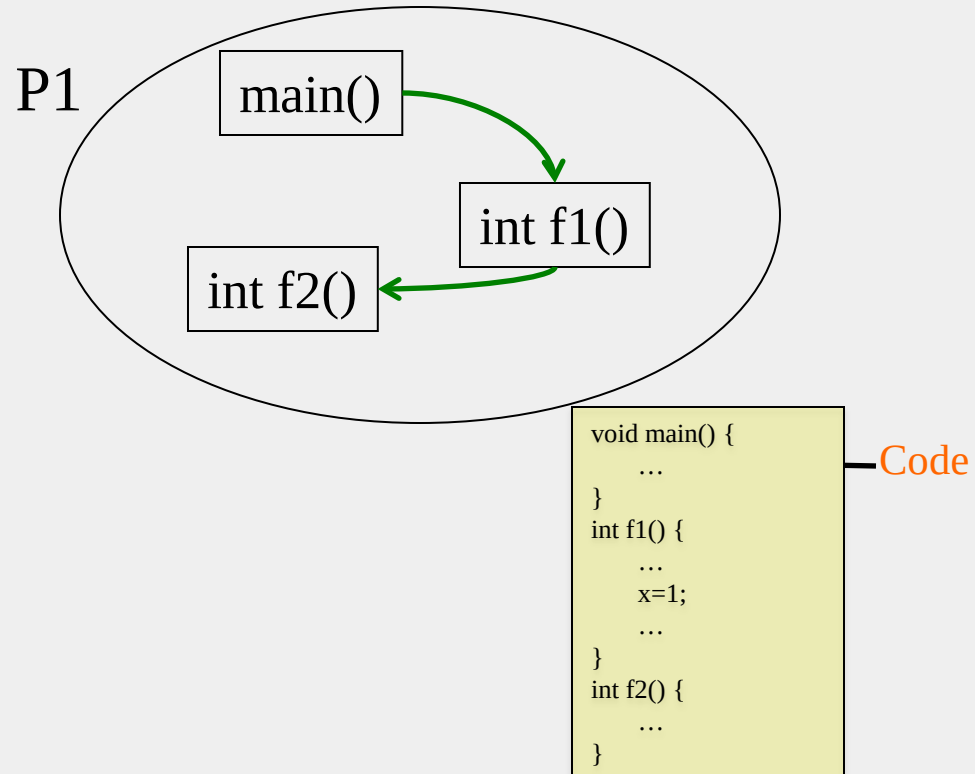
BASIC DATASTRUCTURES: STACK

- Stack: First-in *Last-out* datastructure

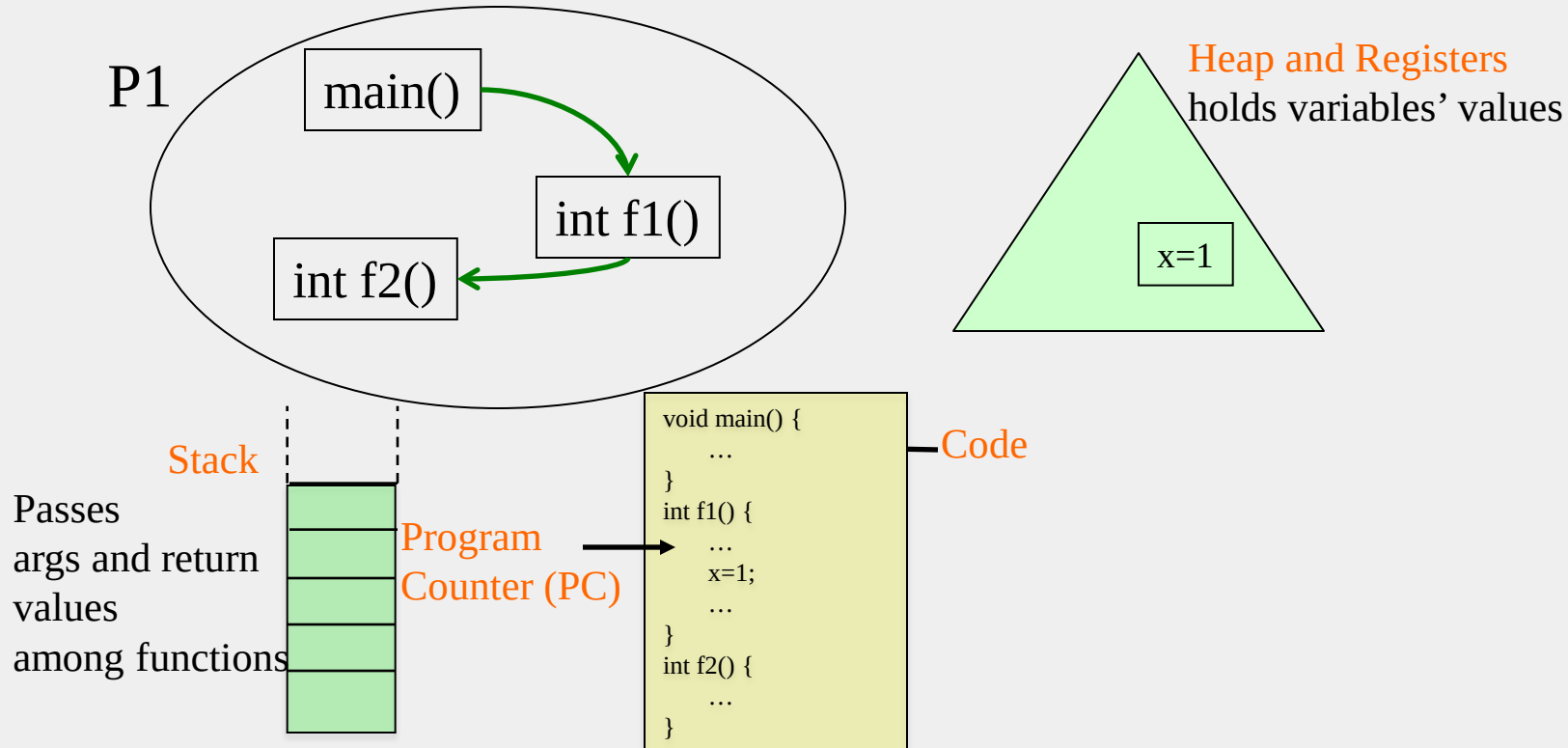


- Insert (Push) 9: goes to top
- Remove (Pop): gets 9
- Pop: gets 3
- Next pop: gets 5 (and so on)

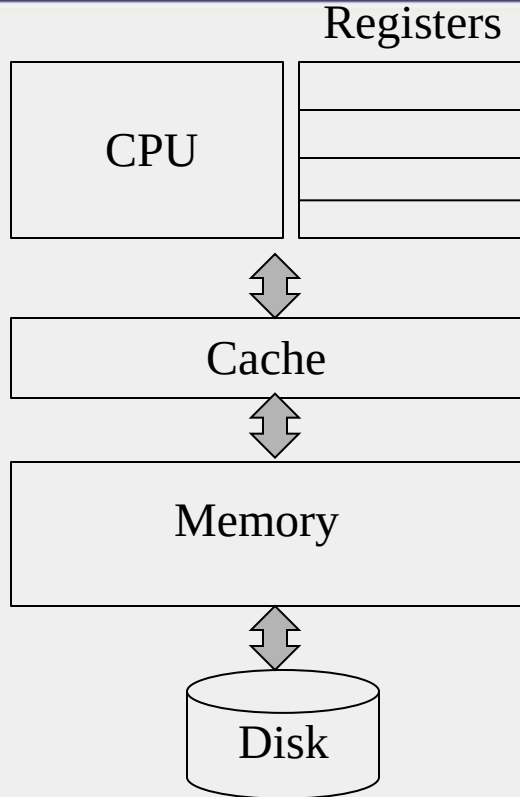
II. PROCESS = A PROGRAM IN ACTION



INSIDE A PROCESS



III. COMPUTER ARCHITECTURE (SIMPLIFIED)



COMPUTER ARCHITECTURE (2)

- A program you write (C++/Java etc.) gets compiled to low-level machine instructions
 - Stored in file system on disk
- CPU loads instructions in batches into memory (and cache, and registers)
- As it executes each instruction, CPU loads data for instruction into memory (and cache, and registers)
 - And does any necessary stores into memory
- Memory can also be flushed to disk
- This is a highly simplified picture!
 - (but works for now)

IV. BIG O() NOTATION

- One of the most basic ways of analyzing algorithms
- Describes *upper bound* on behavior of algorithm as some variable is scaled (increased) to infinity
- Analyzes run-time (or another performance metric)
- Worst-case performance

BIG O() NOTATION: INFORMAL DEFINITION

- “An algorithm A is $O(\text{foo})$ ”

Means

- “Algorithm A takes $< c * \text{foo}$ time to complete, for some constant c , beyond some input size N ”
- Usually, foo is a function of input size N
 - e.g., an algorithm is $O(N)$
 - e.g., an algorithm is $O(N^2)$
- We don't state the constants in Big $O()$ notation

BIG O() NOTATION: EXAMPLE 1

- “Searching for an element in an unsorted list is $O(N)$, where N = size of list”
- Have to iterate through list
- Worst-case performance is when that element is not there in the list, or is the last one in the list
- Thus involves N operations
- Number of operations $< c * N$, where $c=2$.

BIG O() NOTATION: EXAMPLE 2

- “Insertion sorting of an unsorted list is $O(N^2)$, where N = size of list”
- Insertion sort Algorithm:
 - Create new empty list
 - For each element in unsorted list
 - Insert element into sorted list at appropriate position
- First element takes 1 operation to insert
- Second element takes (in worst case) 2 operations to insert
- i -th element takes i operations to insert
- Total time = $1+2+3+\dots+N = N(N+1)/2 < 1*N^2$

V. BASIC PROBABILITY

- **Set**=collection of thingies
 - S ="Set of all humans who live in the world"
- **Subset**=collection of thingies that is part of a larger set
 - S_2 ="Set of all humans who live in Europe"
 - S_2 is a subset of S

BASIC PROBABILITY

- Any event has a probability of happening
- If you wake up at a random hour of the day, what is the probability of the event that the time is between 10am and 11am?
- There are 24 hours in a day
 - Set of hours contains 24 elements: 12am, 1am, 2am, ... 10am, 11am, ...11pm
- You pick one hour at random
- Probability you pick 10am = $1/24$

MULTIPLYING PROBABILITIES

- E1 is an event
- E2 is an event
- E1 and E2 are independent of each other
- Then: $\text{Prob}(\text{E1 AND E2}) = \text{Prob}(\text{E1}) * \text{Prob}(\text{E2})$
- You have three shirts: blue, green, red
- You wake up at a random hour and blindly pick a shirt
- $\text{Prob}(\text{You woke up between 10am and 11am AND that you're wearing a green shirt}) = (1/24) * (1/3) = 1/72$
- But beware: can't multiply probabilities if events are dependent (i.e., influence each other)!

ADDING PROBABILITIES

- E1 is an event
- E2 is an event
- Then:

$$\text{Prob}(E1 \text{ OR } E2) = \text{Prob}(E1) + \text{Prob}(E2) - \text{Prob}(E1 \text{ AND } E2)$$

- If you don't know $\text{Prob}(E1 \text{ AND } E2)$, then you can write

$$\text{Prob}(E1 \text{ OR } E2) \leq \text{Prob}(E1) + \text{Prob}(E2)$$

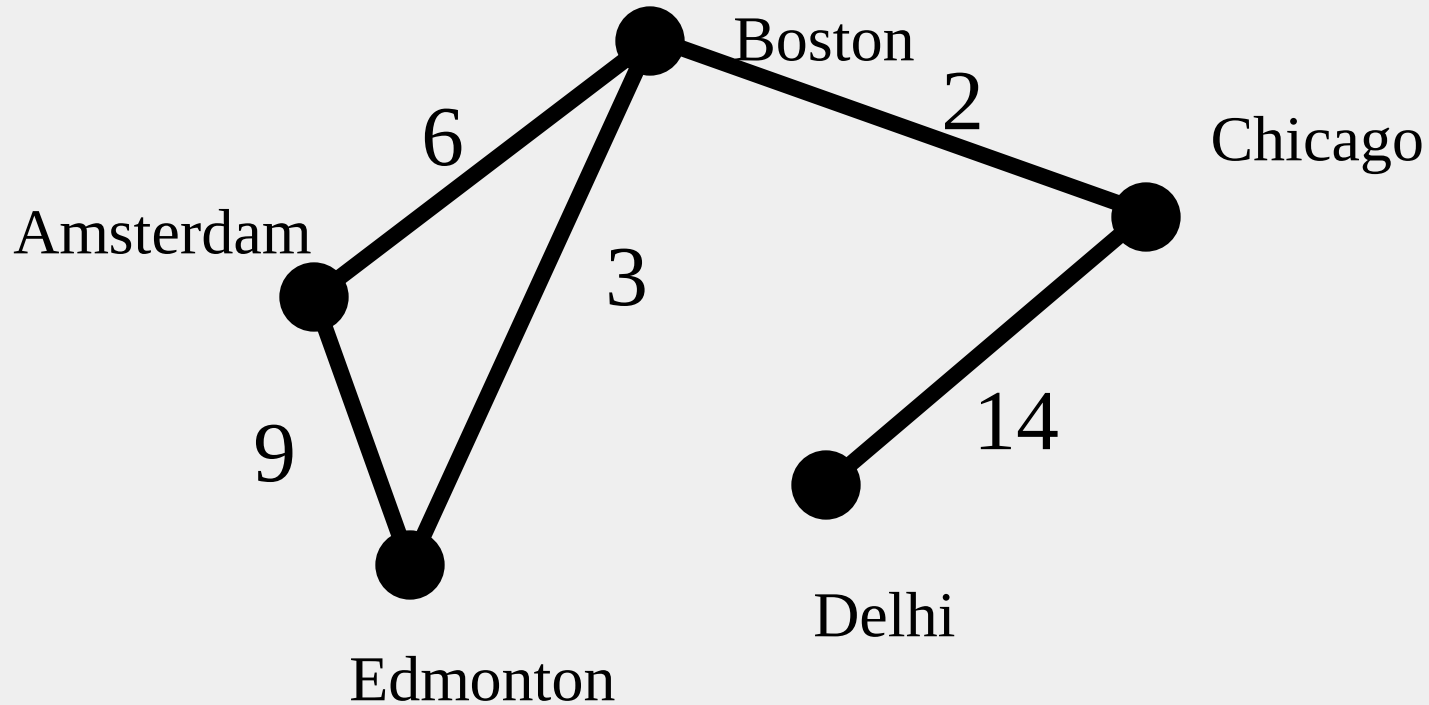
VI. DNS

- DNS = Domain Name System
- Collection of servers, throughout the world
- Input to DNS: a domain name, e.g., coursera.org
 - Domain name is a name, a human-readable string that uniquely identifies the object
 - Can be derived from the URL (e.g., <https://class.coursera.org/foo>)

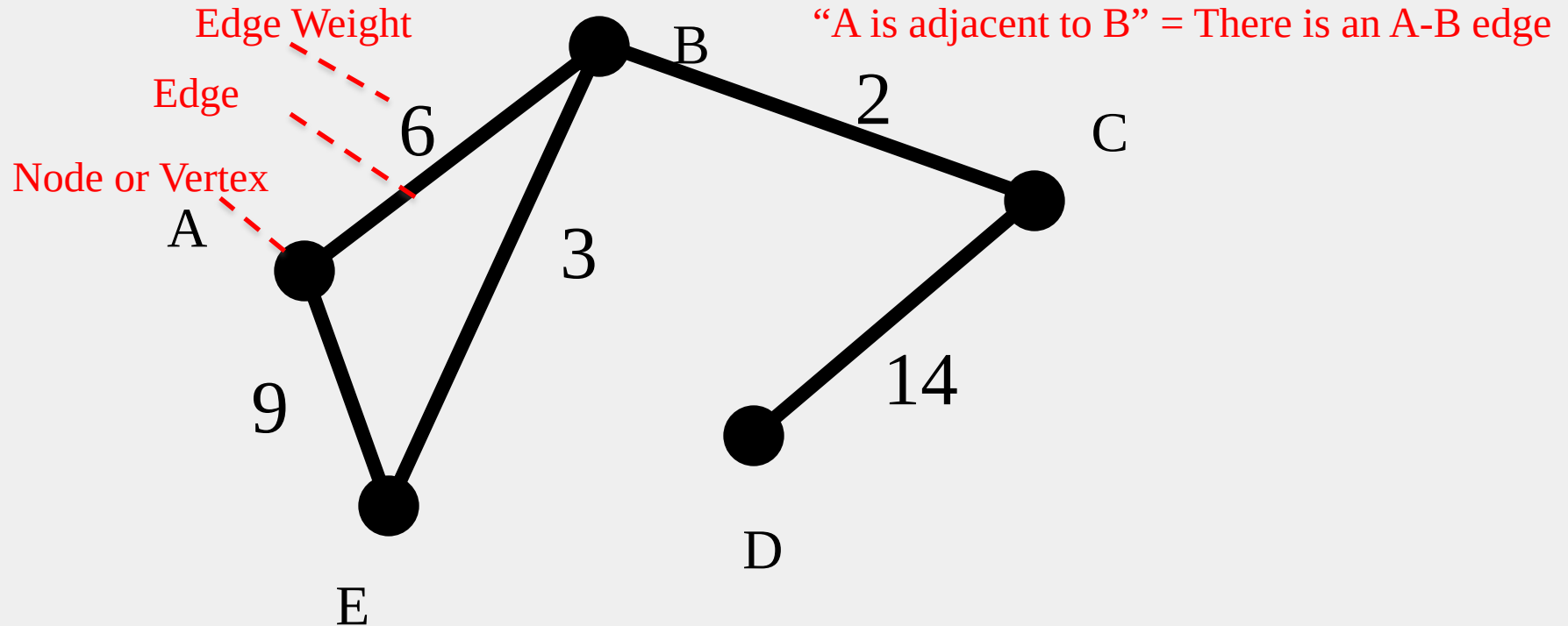
DNS (2)

- Output from DNS: IP address of a web server (that hosts that content)
 - IP address is an ID, a unique string pointing to the object. May not be human readable.
- IP address may refer to either
 - Web server actually hosting that content, or
 - An indirect server, e.g., a CDN (content distribution network) server, e.g., from Akamai

VII. GRAPHS



GRAPHS (2)



WHAT WE COVERED

- I. Basic datastructures
- II. Processes
- III. Computer Architecture
- IV. $O()$ notation
- V. Basic probability
- VI. Miscellaneous